# Tablet Coating

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The first coated pills were used by **Rhazes** (850-923). Sugar coating of pills was extensively developed by Frenchmen. Then a dramatic change was observed in coating when Abbot laboratories marketed first film coated tablet. Dr. Dale Wurster develop air suspension coating technique in 1950s.

**Reasons for Coating**

Different dosage forms may need different kind of coating formulation, technique and process. Therefore a formulation development scientist has to understand the critical aspects associated with each case. Different dosage forms which can be coated are : tablets, capsules, pellets, granules, particles and powder. All the above mentioned dosage forms may need coating for different reasons such as :

1. **Change in appearance:** To impart colour for easy identification during manufacture, dispensing, in use by patient and brand image building.
2. **To eliminate dust generation**: To reduce handling problems and to reduce dust induced toxicity.
3. **Taste masking**: Mask the bitter or unpleasant taste.
4. **Odour masking:** Mask the unpleasant odour of active ingredients like vitamins, antibiotics etc.

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5. **Isolation of incompatible materials:** Some of the ingredients may be incompatible to each other, and these can be separated by putting a barrier coating in between them.

6. **Protection from environmental conditions** : Some of the ingredients may not be stable in the presence of moisture, light, oxygen etc. The product stability can be improved by coating.

7. **Change in release characteristics:** Drug release profile from the dosage form can be tailored by coating techniques for example – delayed release (by enteric coating), extended release (by semi permeable membrane coating or mixing of pellets which are coated to various degree or with different coating materials).

8. **To improve appearance:** often where the core tablet is of poor colour or shows mottling.



Three basic components are critical in Tablet coating.

1.Tablet properties

2.Coating process

1. Coating Equipment
2. Parameters of the coating process
3. Facility and ancillary equipment
4. Automation in coating process.

3.Coating compositions

**1. Tablet Properties**

Tablets that are to be coated must possess the proper physical characteristics. In the coating process, the tablets roll in a coating pan or cascade in the air stream as the coating solution is applied. The tablets must not chip or break.. Also the tablets are exposed to elevated temperature and humidity during the coating. The tablet surface which are brittle gets soften in the presence of heat or effected by coating solutions tends to become rough in the early phase of the coating process are unacceptable for coating, especially for film coating which deposits thin layer of coating, while sugar coating can fill the surface imperfection but upto some extent because of their high solid contents. The tablets must be robust enough to withstand these conditions.

The following must be considered while designing tablets which are to be coated.

**1. Tablet Hardness:** tablet breakage is typically seen when the hardness is inadequate a rule of thumb is the tablets to be coated should have hardness not less than 8 kp (11.2 scu, 80 N)

**2. Tablet Friability:** Friability is more important as that of hardness. This test accurately reflects the stresses that the tablets will when tumbling in a coating pan. The maximum recommended friability for tablets to be coated is 0.5% however for best results the aim should be 0.1%. If the friability is above this limit the tablets may cap, laminate or abrades during coating process.

**3. Tablet core shape:** Tablets may be compressed in variety of shape that include; Round/oblong flat, round/oblong shallow concave, round/oblong standard concave, round/oblong deep concave.

The flat faced tablets have overall best surface hardness buit tend to be brittle at the edges, these tablets are not good for coating because of twinning problem.

The deep concave type good mixing and no twinning problem but they offer the lowest level of surface hardness, so these are also not suitable because of high attrition at crown.



The best compromise between surface hardness, mixing and twinning is offered by the normal or shallow concave shapes and these are recommended for coating.

Generally the flat faced, shallow concave or capsule shaped tablets show greatest erosion at the edges while deep concave show erosion at the crown.

**4. Tablet porosity:** increased tablet porosity can be beneficial in increasing the force of adhesion between the tablet surface and the applied coat. It should be kept in mind increase porosity comes at the sacrifice of tablet hardness.

**5. Tablet core ingredients:** The ingredients both active and inactive can have significant effect on the interaction between the substrate and coating. The most important is the degree of adhesion. The active ingredient is often present in significant amount in the tablet formulation, this will overwhelm the influence on adhesion by other ingredients. If the active content is low and it has negative impact on film adhesion, this may be neutralized by selection of proper excipient.

Lubricants are added to tablet formulations to minimize both die wall friction and punch adhesion, lubricants results in decrease of tablet hardness and coating adhesion, specially when metal (magnesium) stearates are used in combination with MCC and starch based excipients.

Superdisintegrants are valuable materials in tablet formulation due to their ability to reduce disintegration time and increase dissolution rate. However if used indiscriminately, they can exert negative effects on other aspects of tablet quality. High levels of superdisintegrant may result in orange peel effect though the hardness and friability will be well in the limit.

**Coating Process:**

Tablet coating is the application of coating composition to a moving bed of the tablets with concurrent use of heated air to facilitate evaporation of the solvent.

**Equipment:** The process and equipment must provide;

1. Distribution of the liquid coating formulation over the whole of the available tablet surface (ladling, Spraying).
2. Continuous mixing of the tablet load in order to achieve an evenly coated product (rotation).
3. Continuous drying to solidify the film quickly (hot air).
4. Removal of solvent vapour (plus dust generated, plus used drying air and atomizing air).



**Equipment:**

Three types of coating equipments are used.

Standard (conventional) coating pan

Perforated coating pan

Fluidized bed (air suspension) coater

**1. Standard (conventional) coating pan**

The standard pan is 8 – 60 inches in diameter and is rotated on its horizontal axis by a motor. Heated air is directed into the pan an duct and exhausted out by means of another duct positioned over the tablet be in front of the pan. Coating solution is applied either through ladling or spraying.

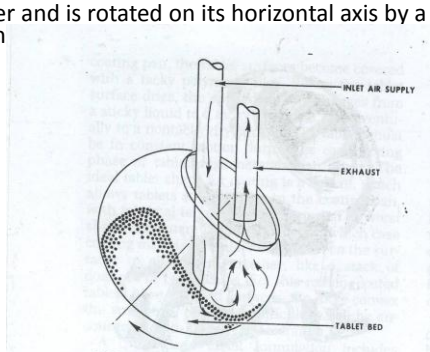
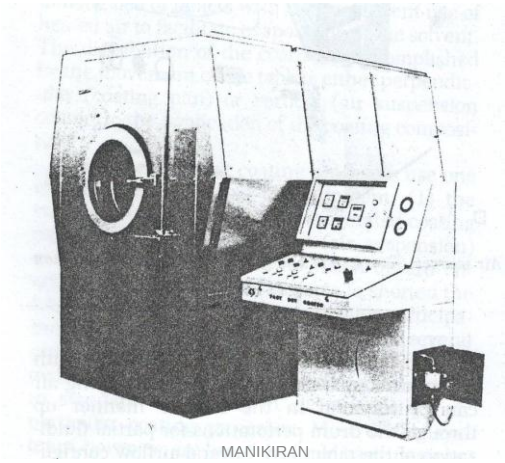



FIG. 12-5. *Diagram of standard coating pans.*

- **Pellegrini pan**

Pellegrini system has baffled pan and a diffuser that distributes the drying air over the tablet bed surface.



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- **Immersion sword pan**

In this system the drying air is introduced through a perforated metal sword device that is immersed in the tablet bed. The drying air flows upward from the sword through the tablet bed thus providing efficient drying of the wetted tablets.

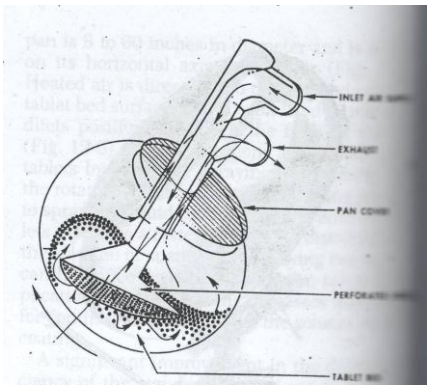


FIG. 12-7. *Simplified diagram of Glatt immersion system.*

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- **Immersion Tube pan**

In this system a tube is immersed in the tablet bed. The tube delivers the heated air, a spray nozzle is built in the tip of the tube to deliver the coating composition. The coating solution and dry air introduced at the same time.

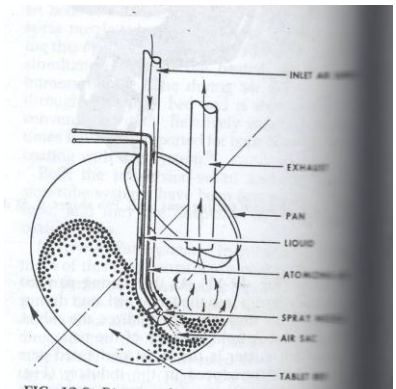



FIG. 12-8. *Diagram of immersion-tube system* (Demmer et al.[5])



- **Perforated pan system**

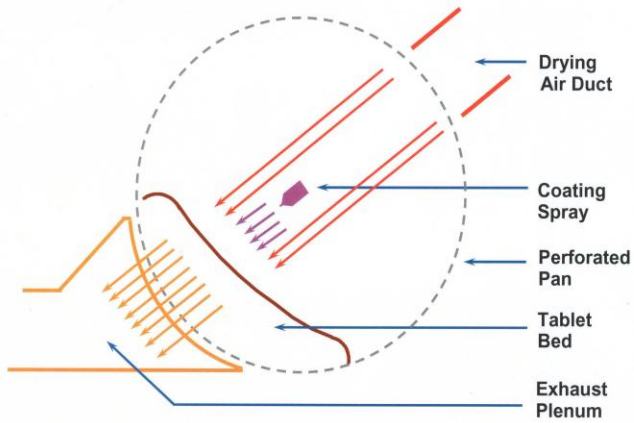
This system make use of perforated or partially perforated drum (pan) that rotates on its horizontal axis in an enclosed housing. This type of coating system are more efficient in drying as compared to the conventional system. With such system mark decrease in coating time can be achieved. The current systems available include;

1. Manesty Accela Cota
2. Freund Hi-Caoter
3. Driacoater
4. Glatt Coater

## SIDE-VENTED COATING PAN

FIG 10

### Principle of operation

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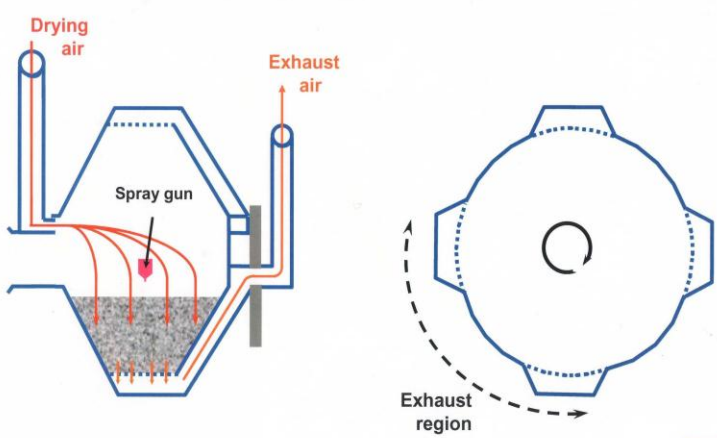
## SIDE-VENTED COATING PAN

FIG 11

### Freund Hi Coater - schematic

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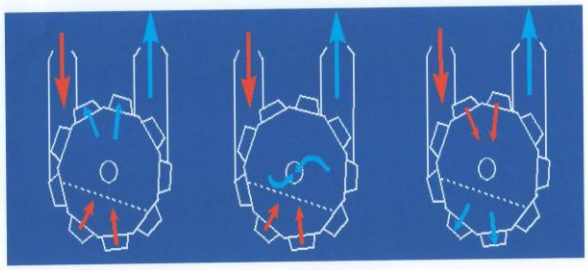
## SIDE-VENTED COATING PAN

FIG 12

### Driam - Driacoater - schematic



1. Reverse air flow Type A
2. Reverse air flow Type B
3. Direct air flow

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- **Fluidized bed (air suspension) coater**

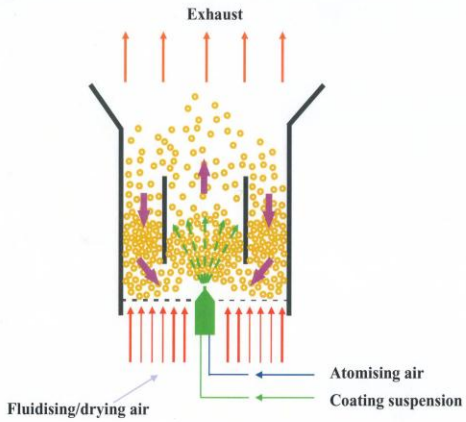
This system uses columnar chamber through which high volume heated air is supplied from the bottom to fluidize the tablet bed, providing the medium for drying and the energy for mixing. These are most efficient drying and mixing equipment. These are not commonly used for tablets on large scale high attrition the tablets chip, breaks, abrades out. This system is commonly applied for particulate coating (microencapsultion and pellet coating).

The coating solution is either applied from the bottom (Wurster) or from the top of the chamber on to the tablet bed.

# FLUID BED COATERS

FIG 13

## Wurster column

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- **Spraying Application Systems (spray gun):**

Two systems are used to apply atomized spray of coating composition onto the tablets.

1. High pressure air less system
2. Low pressure air atomized system

1. **High pressure air less system**

In this system the liquid is pumped at high pressure (250-3000 psig) through a small orifice (0.009-0.20 inch) in the fluid nozzle. The degree of atomization and the spray rate are controlled by the fluid pressure, orifice size, and viscosity of the liquid.

The disadvantage of the system is that,

- Because of the small orifice the suspension may block it therefore the coating suspension may be finely milled or filtered.
- the airless system doesn't provide independent control of suspension application rate, degree of atomization and spray fan width.

2. **Low pressure air atomized system**

In this system liquid is pumped through larger orifice (0.020 – 0.06 inch) at relatively low pressure (5-50 psig). Low pressure of about 10-100 psig at the atomizer tip contact the liquid and disperse it in finely divided spray. In this system atomization can be controlled independently of the operating pressure, also fan width can be controlled.

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# DISTRIBUTION

FIG 1

## Distribution - airless spray gun

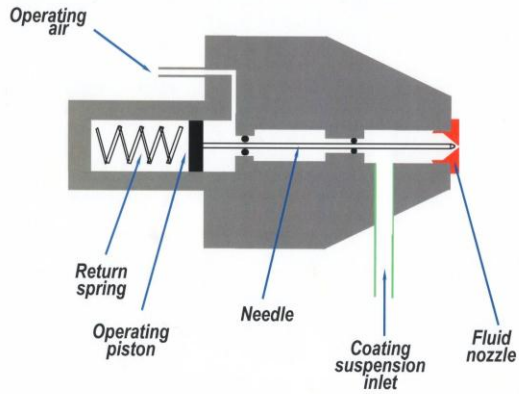


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FIG 2

## Air-atomised spray gun – 3 port

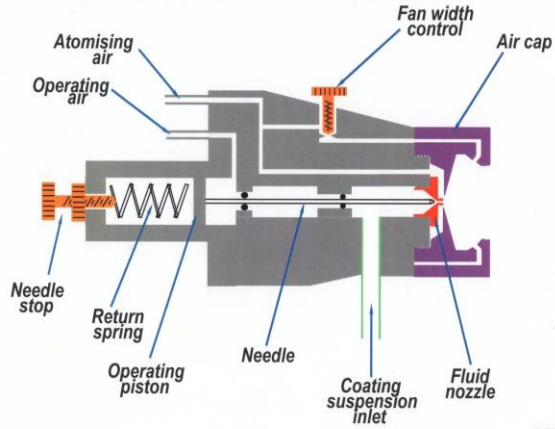


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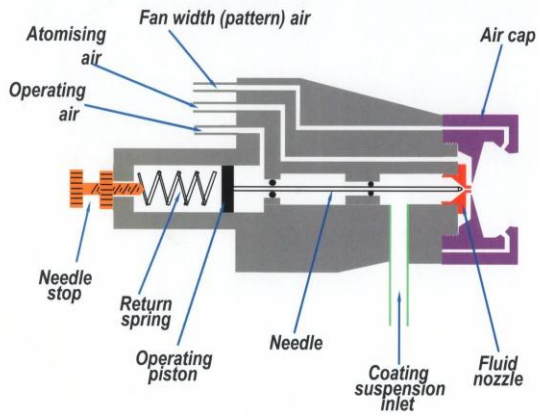
FIG 3

## Air-atomised spray gun – 4 port

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**Coating Parameters**

During coating the tablets move through an application zone in which a portion of the tablets receive some coating. Most of the time tablets are in drying mode moving away from the application zone and recycled repeatedly through the application zone. In the coating operation (continuous) an equilibrium is maintained between coating composition application rate and the rate of evaporation of the solvent. Deviation from this equilibrium results in serious coating problems. These two parameters can be adjusted by;

- **Spray Gun position:** 6-18 inches from the bed, $45^0$ angle to the tablet bed. Not overlapping fan width nor to far.
- **Effect of pan load:** under loading the pan will result in situation where the tablet does not cover fully the exhaust plenum, and the majority of the drying air stream will bypass the tablet bed and drying efficiency will be low.
- **Air volume and temperature:** Drying is controlled by the quantity and temperature of the drying hot air, and the quantity of exhaust air. It is important to balance the inlet and exhaust air flow rates such that there is slight negative pressure in the chamber. Also the temperature difference between the inlet and exhaust must be with in 20-30oC. It is important to monitor the following three temperatures.

**1.** Inlet Air temperature

2. Tablet Bed temperature (most critical)

3. Exhaust air temperature

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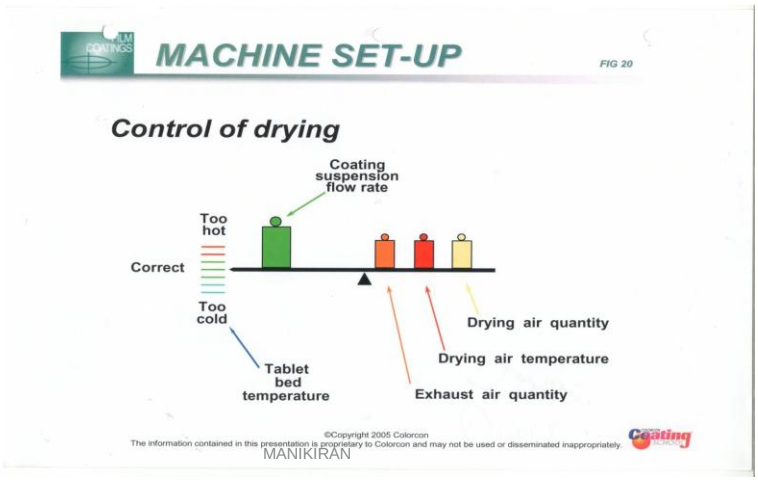
**4. Coating composition:** coating composition contains the ingredients that are to be applied on the surface of tablet and the solvents , which act as carrier for the ingredients. These solvents are not required in the final product and must be removed. A balance must be established between the coating composition flow rate and the three drying variables (drying air temperature/quantity and exhaust air quantity).

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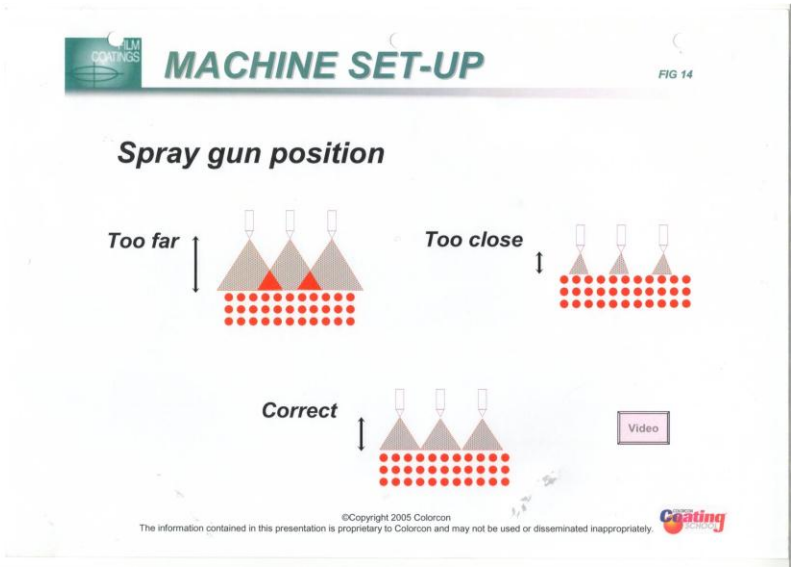


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**Types of coating**

**1.Sugar coating**

**2.Film Coating**

1. **Conventional film coating**
2. **Enteric film coating**

Film coating on the basis of solvent system

1.Organic coating

2.Aqueous coating

**Sugar coating**

This involves several steps, the duration of which ranges from few hours to few days. The quality of coating depends upon the skill of the operator specially in ladling type of solution application. The sugar coating results in elegant highly glossed finished tablets. Following steps are involved in sugar coating.

1. Sealing
2. Subcoating
3. Syruping (smoothing)
4.Finishing
5.Polishing



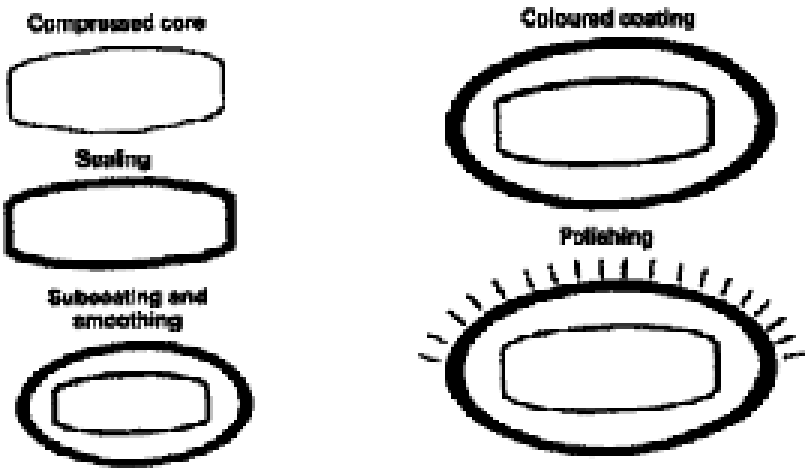


Fig. 3.1 The stages in sugar coating.

**1. Seal coating (sealing)**

To prevent moisture penetration into the tablet core a seal coat is applied. This is specially required in pan-ladling process, to avoid overwetting of the tablet bed. Without seal coat the tablets will absorb sufficient moisture to be softened or disintegrate , effecting the physical appearance. Shellac is the most commonly applied sealant agent, but it offers the problem of increase the disintegration and dissolution time on aging because of polymerization of shellac. Zein is also used as sealant, it has no problem as associated with the shellac. The process description is

- Pan Speed 10 rpm
- Supply air temperature 30oC
- 3 application of 800 ml of sealant solution is applied and the tablets are allowed to dry for 15-20 minutes between the application. If the tablets become tacky at any time apply sufficient talc to avoid sticking.

**2. Sub Coating**

This is applied to round the edges and build up the tablet size. Sugar coating results in the increase of tablet weight by 50-100%. Usually gelatin/acacia solution is used along subcoating powder. The process description is



- Pan speed 10 rpm
- Turn heart and inlet air off. Use exhaust air only
- apply 3-9 coats. Use 1-2 liters for first coat and then reduced the quantity accordingly to obtain the required weight and round edges.
- Allow the tablets to dry at least for 20 minutes after each application. Dust with subcoating powder at the end.
- After the last coat, jog the pan for at least 2-4 hours to ensure complete dryness.

**3. Syrup (smoothing/color) coating**

The purpose of this step is to cover and fill the imperfection in the tablet surface caused by subcoating, this is the most technical step as it requires the most skill. This step usually involves three basic phases.

**1. Grossing syrup (a syrup with subcoating powder dispersed in it):** Process outline is

- Remove excess dust from the pan. Turn on the exhaust air.
- Adjust the temperature to achieve the exhaust air temperature of 45-48oC.
- Pan Speed 15 rpm
- Apply 5-15 coats of grossing syrup.

**2. Heavy syrup:** This solution contain color, maintaining the above parameters apply several coats.

**3. Regular syrup:** The process steps are

• Turn off the heat, reduce the inlet and outlet air.

• apply few coats of regular colored syrup to achieve final smoothness, size and color development.

**4. Finishing**

•With the heat and inlet air supply off , reduced exhaust air and pan speed of 12 rpm apply 3-4 coats of regular colored syrup rapidly.

• shut off the exhaust air and apply last coat of regular syrup without colorant. This will give depth to the color and will enhance the elegance of the coat.

• stop the pan while the tablets are damp and swiftly shift to jog the pan for 15-30 minutes, leave the tablets overnight to dry completely.

**5. Polishing:**

this is the final step, the tablets can be polished in standard coating pan or canvas line polishing pan. The steps involved are

• Pan speed 12 rpm

• inlet air, outlet air turned off

• apply 3-4 coats warm polishing solution (waxes), approximately 300 ml per application.

• Apply the subsequent coat when the solvent is evaporated out.

•Sometimes powder wax are also applied.



**Film Coating**

To reduce sugar coating process time and to reduce the requirement for operator skill, film coating was developed.

Film coating is a technique in which a thin layer/coat of a polymer is deposited over the tablets/particulate. Depending on the coating polymer the types of coating are

1.Normal film coating

2.Enteric film coating

3.Aqueous moisture barrier coating (AMB)

On the basis of solvent use the types are

1.Organic Coating

2.Aqueous Coating

**Materials used in film coating**

A typical film coating formulation is made up of

1. Polymer (film former)

2. Plasticizer

3. Colouring/opacifying agent

4. Solvent

5. Others (surfactants, flavors, sweetening agent, active ingredients and preservatives.)

An ideal coating material should have the following properties;

- Solubility in solvent of choice for coating preparation
- Solubility required for the intended use
- Capacity to produce an elegant looking product
- Stability in the presence of heat, light, moisture, air and substrate being coated
- Odorless, colorless and tasteless
- compatibility with other ingredients
- Non toxic or no pharmacologic activity
- Ease of application
- Resistant to cracking
- No bridging or filling formation
- Ease of printing on high speed machines

**Polymers/Film formers**

The function of the polymer is to provide main structure and basic physical and chemical properties to the coating.

Polymer viscosity is very important specially in aqueous coating we need to minimize the water concentration, it is to shorten the process time and to minimize product exposure to the moisture (moisture sensitive product). But the coating composition with viscosity above 500 cps are difficult to atomize and will not produce smooth product. Therefore polymers with low viscosity are preferred.



## Nonenteric polymers

The commonly used polymers are

**1. Hydroxypropyl methylcellulose (HPMC)**

Commercially it is available in different viscosity grades This polymer is widely used in air suspension and pan spray coating. The properties are

1. Soluble in GI fluid, organic and aqueous solvent system
2. Noninterference with tablet disintegration and drug availability
3. Flexibility
4. Chip resistant
5. No odor or taste
6. Stability in presence of light, heat, air and reasonable amount of moisture
7. Incorporation of colour and other additives with out difficulty

If it is used alone then it may results in bridging and filling, so it may be used in combination or proper plasticizer may be used.

**2. Methyl hydroxyethylcellulose**

This polymer has also different viscosity grades, it has similar properties as that of HPMC but it is soluble in few organic solvents, which has restricted its use.

**3. Ethylcellulose**

Depending on the ethoxy substitution different viscosity grades are available commercially. This material is water and GI fluid insoluble, therefore can not be used alone for coating. It is used in combination with water soluble polymer e.g. HPMC. These combination are a used for sustained release coating. The properties include

1. Soluble in wide variety of organic solvents
2. Non toxic
3. Tasteless, odorless and colorless
4. stable at environmental conditions

Unplasticized ethyl cellulose coats are brittle.

**4. Hydroxypropylcellulose**

1.It is soluble in water below 40$^{o}$C, GI fluid and organic solvents

2. it is very tacky and yield very flexible film, it can not be used alone

3. it is good for sub coat but not for color or gloss coat

**5. Povidone (PVP)**

1.It is available in four viscosity grades i.e. K-15, K-30, K-60 and K-90, the average molecular weights are 10,000, 40,000, 160,000 and 360,000 respectively.

2.It is soluble in water, GI fluid and variety of organic solvents.



3. Povidone films are clear, hard and glossy
4. Povidone is soluble in both acidic and intestinal media, it can be cross linked with other materials to produce enteric coating material.
5. povidone is used in coating composition to increase the dispersion of color.

**3. Sodium Carboxymethylcellulose**

1. It is available in low, medium ,high and extra high viscosity grades
2. It can easily be dispersed in water to form colloidal solution, but is insoluble in most organic solvents.
3. the film formed is brittle but adheres well to tablets.

**7. Acrylate polymers**

Acrylate polymers are marketed under the trademark of **Eudragrit.** Eudragrit E(dimethylaminoethyl methacrylate and other neutral methacrylic acid esters) is the only polymer of this group which is freely soluble in gastric fluid (acidic media).

**8. Polyethylene Glycols**

## Enteric Polymers

Enteric coating polymers are those substances which resists the gastric pH (acidic) and get dissolves in intestinal fluid (alkaline). The reasons for enteric coating are

- To protect acid labile drugs from gastric fluid e.g. enzymes and certain antibiotics
- To prevent gastric distress or nausea e.g. sodium salicylate
- To deliver drug to intestine for local action
- To deliver drugs that are optimally absorbed in the small intestine
- To provide a delayed release component for repeat action tablets.

An ideal enteric coating polymer should have the following properties

1. Resistance to gastric fluids (acidic pH)
2. Ready susceptibility to or permeability to intestinal fluid (alkaline pH)

- Compatibility with other ingredients
- Non toxic or no pharmacologic activity
- Ease of application
- Formation of continuous film
- stability alone and in coating solution. The film should not change with aging
- Ease of printing on high speed machines



1. **Cellulose Acetate Phthalate (CAP)**

CAP is widely used but it has major disadvantage that it dissolves above pH 6, thus delaying the drug release as the ideal material may dissolve around pH 5.

It is also hygroscopic and relatively permeable to gastric fluid. The film formed is brittle thus required plasticizer. It is available under the trademark of **Aquateric** from FMC corporation.

**2. Acrylate polymers.**

Two commercially available acrylates are

• Eudragrit L (soluble at pH 6), Eudragrit S (soluble at pH 7)

**3. HPMC Phthalate**

Three grades are available

• HPMCP 55 (HP 50), HPMCP 50 (HP 55) and HPMCP 55S (HP 55S)

These polymers dissolves at lower pH (at 5-5.5) than CAP and acrylates. Thus resulting in higher bioavailability of some specific drugs.

**4. Polyvinyl Acetate Phthalate (PVAP)**

It is similar to HP 55 in stability and pH dependant solubility.

**Plasticizer**

The function of plasticizer is to modify the basic mechanical properties of the polymer. Plasticizers have high affinity for the polymer they are also called nonvolatile solvents.

There are two techniques to modify the plasticity of the film former;

1.Internal plasticizing

2.External plasticizing

In the former technique chemical modification is brought in the polymer which alters the physical properties of the polymer i.e. elastic modulus. In later technique other substances are used as plasticizer in the formulation. When the plasticizer is used in correct concentration it imparts flexibility by relieving the molecular rigidity. Commonly 10% of polymer concentration is used. Examples include

1.Castor Oil

2.PEG 200 and 400

3.Propylene glycol

4.Glycerin

5.Polysorbates (tweens)

6.Sorbitan esters (spans)



**Coloring and opacifying agent**

The function of these ingredients is to enhance the product quality. They provide

1.Product identification

2.Protect the core from light and moisture

3.They increase the solid concentration with any impact on viscosity thus reducing the drying time. They are either soluble or form fine suspension in the solvent system. For uniform distribution the particle size must be < 10 microns. The most common colorants used are FD&C or D&C certified, these are either dyes or lakes of dyes. Examples include iron oxide, anthrocyanins, caramel, carotenoids, chlorophyll, indigo, flavones, turmeric acid and carminic acid.

Opacifiers are used to give more pastel color and increase film coverage. These can provide white coat or mask the color of the tablet core. These are mostly inorganic material. The substances employed are

1.Titanium dioxide (Most Common)

2.Talc

3.Aluminum silicate

4.Magnesium carbonate

5.Calcium sulfate

6.Aluminum hydroxide

**Solvents**

The function of the solvent is to dissolve or disperse the polymers and other additives and transfer them to the surface of substrate (core) the ideal characteristics are

1.It should either dissolve or disperse the polymer system and other ingredients.

2.Small concentration of polymer (2-10%) should not high viscous solution

3.It should be colorless, tasteless, odorless, inexpensive, inert and noninflammable.

4.Should have rapid drying rate

5.Should not have environmental impact.

The most widely solvents used either alone are in combination are water, ethanol, methanol, isopropyl alcohol, chloroform , acetone , methyl ethyl ketone and methylene chloride.



## Problems and remedies for tablet coating

**Blistering**

**Definition:** It is local detachment of film from the substrate forming blister.

**Reason:** Entrapment of gases in or underneath the film due to overheating either during spraying or at the end of the coating run.

| Sr. No | CAUSE | REMEDY |
|---|---|---|
| 1. | Effect of temperature on the strength, elasticity and adhesion of the film. | Use mild drying condition. |

**Chipping**

**Definition:** It is defect where the film becomes chipped and dented, usually at the edges of the tablet.

**Reason:** Decrease in fluidizing air or speed of rotation of the drum in pan coating.

| Sr. No | CAUSE | REMEDY |
|---|---|---|
| 1. | High degree of attrition associated with the coating process. | Increase hardness of the film by increasing the molecular weight grade of polymer. |

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**Cratering**

**Definition:** It is defect of film coating whereby volcanic-like craters appears exposing the tablet surface.

**Reason:** The coating solution penetrates the surface of the tablet, often at the crown where the surface is more porous, causing localized disintegration of the core and disruption of the coating.

| Sr. No. | CAUSES | REMEDIES |
|---|---|---|
| 1. | Inefficient drying. | Use efficient and optimum drying conditions. |
| 2. | Higher rate of application of coating solution. | Increase viscosity of coating solution to decrease spray application rate. |

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**Sticking and Picking**

**Definition:** It is defect where isolated areas of film are pulled away from the surface when the tablet sticks together or to the coating pan and then detached from one another or from the pan and piece of film get remained to the pan or the other tablet exposing the core.

***Reason:*** Conditions similar to cratering that produces an overly wet tablet bed where adjacent tablets can stick together and then break apart.

| Sr.No | CAUSE | REMEDY |
|---|---|---|
| 1. | Inefficient drying. | Use optimum and efficient drying conditions or increase the inlet air temperature. |
| 2. | Higher rate of application of coating solution | Decrease the rate of application of coating solution by increasing viscosity of coating solution. |

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**Pitting**

**Definition:** It is defect whereby pits occur in the surface of a tablet core without any visible disruption of the film coating.

**Reason:** Temperature of the tablet core is greater than the melting point of the materials used in the tablet formulation.

| **Sr. No** | **CAUSE** | **REMEDY** |
|---|---|---|
| 1. | Inappropriate drying (inlet air ) temperature | Dispensing with preheating procedures at the initiation of coating and modifying the drying (inlet air) temperature such that the temperature of the tablet core is not greater than the melting point of the batch of additives used. |

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**Blooming/Hazing/Dull Film**

**Definition:** It is defect where coating becomes dull immediately or after prolonged storage at high temperatures.

***Reason:*** It is due to collection on the surface of low molecular weight ingredients included in the coating formulation. In most circumstances the ingredient will be plasticizer.

| Sr. No. | CAUSE | REMEDY |
|---|---|---|
| 1. | High concentration and low molecular weight of plasticizer. | Decrease plasticizer concentration and increase molecular weight of plasticizer. |
| 2. | High processing temperature, especially when using cellulosic polymer | Appropriate drying temperature |

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**Blushing**

**Definition:** It is defect best described as whitish specks or haziness in the film.

***Reason:*** It is thought to be due to precipitated polymer exacerbated by the use of high coating temperature at or above the thermal gelation temperature of the polymers.

| Sr. No. | CAUSES | REMEDIES |
|---|---|---|
| 1. | High coating temperature | Decrease the drying air temperature |
| 2. | Use of sorbitol in formulation which causes largest fall in the thermal gelation temperature of the Hydroxy Propyl Cellulose, Hydroxy Propyl Methyl Cellulose, Methyl Cellulose and Cellulose ethers. | Avoid use of sorbitol with Hydroxy Propyl Cellulose, Hydroxy Propyl Methyl Cellulose, Methyl Cellulose and Cellulose ethers. |

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**Colour variation**
**Definition:** A defect which involves variation in colour of the film.

***Reason:*** Alteration of the frequency and duration of appearance of tablets in the spray zone or the size/shape of the spray zone.

| Sr. No. | CAUSE | REMEDY |
|---|---|---|
| 1. | Improper mixing, uneven spray pattern, insufficient coating, migration of soluble dyes-plasticizers and other additives during drying. | Go for geometric mixing, reformulation with different plasticizers and additives or use mild drying conditions. |

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**Orange peel/Roughness**

**Definition:** It is surface defect resulting in the film being rough and nonglossy. Appearance is similar to that of an orange.

***Reason:*** Inadequate spreading of the coating solution before drying.

| Sr. No. | CAUSES | REMEDIES |
|---|---|---|
| 1. | Rapid Drying | Use mild drying conditions |
| 2. | High solution viscosity | Use additional solvents to decrease viscosity of solution. |
| 3 | Indiscriminate use of superdisintegrant | Correct quantity, decrease flow rate |

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**Cracking/Splitting**

**Definition:** It is defect in which the film either cracks across the crown of the tablet (cracking) or splits around the edges of the tablet (Splitting)

**Reason:** Internal stress in the film exceeds tensile strength of the film.

| Sr. No. | CAUSE | REMEDY |
|---|---|---|
| 1. | Use of higher molecular weight polymers or polymeric blends. | Use lower molecular weight polymers or polymeric blends. Also adjust plasticizer type and concentration. |



**Bridging and Filling**

**Definition:** Bridging is defect in which the bisects or monogram is obscured. During drying the film may shrinks and pull away from the corners of intagliation or bisects. In filling the bisects or monogram is filled with coating solution.

| Sr. No. | CAUSE | REMEDY |
|---|---|---|
| 1. | Improper or inadequate polymers selection. | Proper plasticizer type and concentration. |
| 2. | Applying to much solution to form thick films | Proper solution application. |
| 3. | To rapid solution application causes deposition of liquid in narrow monogram or bisects results in filling | Reduce the flow rate of coating composition. |